28 The size of each interior angle of a regular polygon is 11 times the size of each exterior angle.

Work out how many sides the polygon has.

Interior + exterior = 180° on a straight line.

Their ratio is 11 : 1, so 12 parts = 180°.

1 mark
Find the exterior angle

One part = 180° ÷ 12 = 15°.

	Exterior	Interior	
×15	1	11	
	15	165	×15

Exterior angles total 360°.

1 mark
Divide 360 by the exterior angle

Number of sides = 360° ÷ 15° = 24.

24

1 mark
Correct number of sides

(Total for Question 28 is 3 marks)